

ANKOMAH AIDOO, NATHANIEL

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PROFILE

Computer Science graduate and current Teaching Assistant at KNUST with research interests at the intersection of Secure Software Engineering and Artificial Intelligence. Equipped with a strong theoretical and practical background in systems architecture, networking, and full-stack software development. Proven track record of academic leadership, undergraduate technical instruction, and applied software engineering.

EDUCATION

Kwame Nkrumah University of Science and Technology (KNUST) Kumasi, Ghana
BSc. Computer Science Jan 2022 – Oct 2025

- **Relevant Coursework:** Introduction to Artificial Intelligence, Discrete Mathematics, Probability & Statistics, Linear & Numerical Algebra, Numerical Methods, Data Structures & Algorithms, Structured Program Design, Object-Oriented Programming (Java), Mobile Application Development, Introduction to Compilers.

Nkawie Senior High School (Technical) Nkawie, Ghana
W.A.S.S.C.E (General Science) Sept 2016 – Sept 2019

SKILLS AND ABILITIES

Programming Languages: Python, JavaScript, Dart, Bash Scripting, Linux Command Line, HTML5, CSS3, SQL.

Frameworks / Libraries: FastAPI, Flutter, Riverpod, React Native, React.js, LangChain, TensorFlow, TFLearn, Pandas, NumPy.

Developer Tools & Cloud: AWS (EC2, VPC, S3, IAM, RDS), Git/GitHub, PostgreSQL, MySQL, ChromaDB, XAMPP.

Soft Skills: Problem-Solving, Cross-Functional Team Leadership, Technical Mentorship, Clear Communication.

WORK EXPERIENCE

Research and Teaching Assistant Kumasi, Ghana
Department of Computer Science, KNUST Nov 2025 – Present

- Facilitated foundational and technical instruction, leading laboratory sessions and supporting assessment activities for over 300 undergraduate Computer Science students.
- Provided research support for departmental projects, contributing to data collection, analysis, and project development.
- Mentored students on programming, data structures, and computer architecture assignments, guiding problem-solving and technical concepts.

Mobile App Developer Kumasi, Ghana
RAPiD-VBP, KNUST Jun 2026 – Aug 2026

- Designed and developed a responsive mobile vector surveillance application featuring custom interactive geospatial maps.
- Implemented a mobile computer vision pipeline to process deep-learning model outputs across five taxonomic levels (verification, stage, genus, species, and sex) while enforcing morphological constraints programmatically.

AWS Certified Cloud Practitioner Trainee Kumasi, Ghana
AmaliTech AWS re/Start Program Sept 2024 – Dec 2024

- Executed intensive training mapping remote resource scheduling and cluster topologies to the AWS Well-Architected Framework.
- Provisioned distributed backend environments utilizing core AWS services, including VPC subnets, EC2 instances, S3 storage buckets, and IAM access controls.

WordPress Developer Accra, Ghana (Remote)
Premier Care GH Jul 2024 – Aug 2024

- Developed a responsive website for a pharmaceutical holding company using WordPress, HTML5, CSS, and XAMPP.
- Showcased the company's marketing, sales, and distribution of health products across modern web browsers and mobile viewports.

Frontend Developer Intern

Center for Applied Research and Innovation in Supply Chain-Africa (CARISCA, KNUST)

Kumasi, Ghana

Nov 2023 – Dec 2023

- Replicated a flip game website and an interactive gaming website landing page.
- Developed JavaScript skills and refined modern HTML5, CSS layout, and client-side styling practices.

PROJECTS

Vector Surveillance and AI Mosquito Classifier

Flutter, Dart, Riverpod, Mobile ML Inference, Google Maps API

- Implemented a mobile computer vision inference pipeline that parses deep learning model outputs across five taxonomic tiers (Verification, Stage, Genus, Species, Sex) and enforces strict morphological state constraints programmatically.

KNUST Students' Pal – Academic Capstone

 [Repository](#)

Tools: Python, FastAPI, LangChain, ChromaDB, OpenAI API, PostgreSQL, React Native

- Engineered a full-stack mobile assistant featuring a vectorized knowledge chatbot, dynamic scheduler, and scholarship tracker.
- Implemented automated data scraping pipelines, LangChain integration, and embedded vector storage (ChromaDB) to power low-latency local queries.

Web Calculator State Management Utility

 [Live Demo](#)

Tools: HTML5, CSS3, JavaScript

- Constructed an isolated web application implementing exact functional array mutations (`reduce`) to preserve arithmetic operand state tracking and enforce real-time division-by-zero validation safety parameters.

Flip Game App

 [Live Demo](#)

Tools: HTML5, CSS3, JavaScript

- Developed a memory card flip game homepage with instructions and a clean user interface featuring interactive card flipping functionality designed to be responsive across different screens.

LEADERSHIP EXPERIENCE

Computer Science Society (CSS, KNUST)

Kumasi, Ghana

President

May 2024 – Sep 2024

- Worked with the executive team to plan and execute events, projects, and initiatives fostering the learning and development of over 2,000 computer science students and enthusiasts.
- Collaborated with external partners and stakeholders to secure funding, resources, and opportunities for the society and its members.
- Led the organization of the department's first computer science symposium, which recounted the evolution of computer science education in Ghana, current industry opportunities, and future prospects.
- Procured a mobile Public Address System with two wireless microphones to augment fixed lecture hall audio infrastructure following increased student enrollment.

Computer Science Society (CSS, KNUST)

Kumasi, Ghana

Vice President

Jan 2024 – May 2024

- Secured 50% academic scholarships for 4 members of the society and established a dedicated scholarship assistance guidance desk.
- Co-led the creation of a learning support system designed to assist 200 students preparing for resit coursework.
- Partnered with Zap Technologies to organize a subsidized mobile app development (React Native) bootcamp for 50 students.
- Established an undergraduate mentorship program where over 700 first-year students were paired with senior colleagues.

ACHIEVEMENTS & AWARDS

- **Winning Team, Engineering Project Competition for Senior High Schools** – Ghana Institution of Engineering (GhIE) Nov 2018

PROFESSIONAL TRAINING & CERTIFICATIONS

- **Fiber Optics Splicing and OTDR Hands-On Training** – JOHVIC Fibertech Solutions Mar 2025
- **AWS Certified Cloud Practitioner (Foundational)** – Amazon Web Services Jan 2025
- **AWS re/Start Graduate** – Amazon Web Services / AmaliTech Dec 2024
- **UITS Student Training in Systems, Networking & Cybersecurity** – KNUST Nov 2023

VOLUNTARY & EXTRA-CURRICULAR ACTIVITIES

- **Content Editor (The Hulede Scholarship Foundation):** Crafts compelling narratives and features highlighting scholarship beneficiaries' community service projects for publication on the foundation's official website and social media platforms.

ACADEMIC REFEREES

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